MASTER STUDY GUIDE: SCIENCE 2 COMPREHENSIVE COMPENDIUM

Secondary School Certificate (SSC) Class 10 • Comprehensive Academic Companion

Executive Study Architecture

This Master Study Guide serves as an authoritative synthesis of the Maharashtra State Board Class 10 Science 2 curriculum. It integrates concise theoretical revision, annotated conceptual diagrams, comparative tables, and comprehensive model solved exemplars aligned with Bloom's Revised Taxonomy.

Module Directory

- Unit I: Central Dogma, Molecular Genetics & Mutation Dynamics (Pages 2–3)
- Unit II: Evidences of Evolution, Chronology & Connecting Links (Pages 4–5)
- Unit III: Human Reproduction, Gametogenesis & Menstrual Cycle (Page 6)
- Unit IV: Conservation Ecology, Biodiversity & Solid Waste Strategies (Page 7)
- Unit V: Green Energy Systems & Electric Power Thermodynamics (Page 8)
- Unit VI: Psychosocial Health, Stress Management & Digital Wellness (Page 9)
- Unit VII: Disaster Management Operational Lifecycle & First Aid Protocols (Page 10)
- Unit VIII: Solved Exemplars 1 – Objective & Correlative Questions (Page 11)
- Unit IX: Solved Exemplars 2 – Scientific Reasons & 2-Mark Explanations (Page 12)
- Unit X: Solved Exemplars 3 – Comparative Distinctions & Tables (Page 13)
- Unit XI: Solved Exemplars 4 – 3-Mark Analytical Justifications (Page 14)
- Unit XII: Solved Exemplars 5 – 5-Mark Comprehensive Case Studies (Page 15)
- Unit XIII: High-Yield Revision Matrix & Examination Success Strategy (Page 16)

UNIT I: CENTRAL DOGMA & PROTEIN SYNTHESIS

Molecular Genetics: DNA Transcription, RNA Translation, and Ribosomal Translocation

1. The Central Dogma Principle

The Central Dogma establishes the unidirectional transmission of biological information encoded in master DNA blueprints into operational cellular protein molecules: DNA -> mRNA (Transcription) -> Polypeptide Protein (Translation).

2. Transcription: mRNA Synthesis

- • Occurs within the cellular nucleus governed by RNA polymerase enzymes.
- • The hydrogen bonds of double-stranded DNA unwind selectively at the promoter site.
- • A single-stranded messenger RNA (mRNA) is synthesized complementary to the DNA template strand.
- • In RNA nucleotide assembly, Uracil (U) substitutes for Thymine (T) opposite Adenine (A).

3. Translation & Triplet Codons

- • Synthesized mRNA departs through nuclear pores into the cytoplasm and binds to ribosomal subunits.
- • The mRNA sequence is deciphered in discrete 3-nitrogenous-base units termed Triplet Codons.
- • Dr. Har Gobind Khorana shared the Nobel Prize (1968) for deciphering amino acid triplet codon mappings.
- • Transfer RNA (tRNA) possessing complementary Anticodons shuttles specific amino acids to the active ribosomal decoding site.

4. Ribosomal Translocation & Polypeptide Elongation

- • Peptide bonds link adjacent amino acids via catalytic activity of ribosomal RNA (rRNA).
- • The ribosome advances sequentially along the mRNA by the exact distance of one triplet codon: Translocation.

UNIT I (Contd.): MUTATION DYNAMICS & SPECIATION

Hugo de Vries Mutation Principle & Mechanisms of Speciation

1. Genetic Mutation Mechanisms

- • A gene is a functional segment of DNA regulating specific biochemical traits.
- • Any sudden accidental modification in nucleotide sequence is termed a Mutation.
- • Minor point mutations may cause hereditary disorders such as Sickle-cell anaemia (substitution of valine for glutamic acid in hemoglobin beta chain).
- • Major chromosomal mutations introduce structural inversions, deletions, or duplications providing raw variations for natural selection.

2. Speciation Drivers

- • Species Definition: A group of morphologically and genetically similar organisms capable of freely interbreeding under natural conditions to produce fertile offspring.
- • Geographical Isolation: Physical barriers (mountain ranges, marine expansion, continental rift) partition a parent population, halting gene flow.
- • Reproductive Isolation: Divergent mating periods, anatomical differences, or courtship rituals prevent interbreeding even if geographical contact is restored.
- • Natural Selection Differential: Divergent environments impose distinctive survival pressures, cementing genomic divergence.

UNIT II: EVIDENCES OF ORGANIC EVOLUTION

Morphological, Anatomical, Vestigial, Palaeontological & Embryological Evidence

1. Comparative Morphology & Anatomy

- • Morphological Evidence: Similarities in facial characteristics, snout orientation, ear pinna, and hair coverage across mammalian lineages suggest descent from common ancestors.
- • Anatomical Evidence: Forelimb of human, foreleg of ox, flipper of whale, and patagium of bat possess radically different functions (grasping, walking, swimming, flying) yet display identical underlying bone configurations (humerus, radius, ulna, carpals, metacarpals, phalanges).

2. Vestigial Organs in Humans

- • Degenerated or functionless structures that were functional in ancestral species.
- • Appendix: Non-functional in humans; fully functional in ruminants for cellulose breakdown.
- • Coccyx (Tail-bone): Remnant of ancestral mammalian vertebral tail.
- • Wisdom Teeth (Third Molars): Redundant due to modern cooked food diets.
- • Auricular Muscles: Non-functional in humans; functional in monkeys, dogs, and horses for ear rotation.

3. Palaeontological Evidence & Carbon Dating

- • Stratigraphic fossil distribution demonstrates ancient invertebrates in deep Paleozoic layers, followed by fishes, amphibians, reptiles in Mesozoic, and birds/mammals in Coenozoic.
- • Willard Libby Carbon Dating: Calculates ratio of radioactive C-14 decay relative to non-radioactive C-12.

UNIT II (Contd.): CONNECTING LINKS & HUMAN EVOLUTION

Living Evolutionary Bridges & The 7-Crore Year Chronology to Homo Sapiens

1. Living Connecting Links

- • Organisms exhibiting morphological and physiological characteristics spanning two distinct phyla or classes.
- • Peripatus: Connecting link between Annelida (segmented body, thin cuticle, parapodia) and Arthropoda (tracheal respiration, open circulatory system).
- • Duck-billed Platypus: Connecting link between Reptiles (oviparous egg-laying, cloaca) and Mammals (mammary glands, hair).
- • Lungfish: Connecting link between Pisces (fins, scales) and Amphibia (pulmonary respiration through lungs).

2. Chronology of Human Evolution

- • 7 Crore Years Ago: Extinction of dinosaurs during the Earth Ice Age; emergence of arboreal Lemur-like ancestors.
- • 4 Crore Years Ago: Aegyptopithecus in Africa lost tails; enlargement of brain and stereoscopic vision.
- • 2 Crore Years Ago: Dryopithecus and Ramapithecus adapted to open grasslands.
- • 40 Lakh Years Ago: Australopithecus exhibited bipedal locomotion and upright posture.
- • 20 Lakh Years Ago: Homo habilis (Handy Man) crafted primitive stone tools.
- • 15 Lakh Years Ago: Homo erectus discovered fire and migrated into Asia and Europe.
- • 1.5 Lakh Years Ago: Neanderthal Man buried dead with ceremonial rituals.
- • 50,000 Years Ago: Cro-Magnon Man developed cave art and advanced tools.
- • 10,000 Years Ago: Modern Homo sapiens initiated organized agriculture and settled civilizations.

UNIT III: HUMAN REPRODUCTIVE SYSTEM & HEALTH

Anatomy, Hormonal Regulation, Menstrual Cycle & Assisted Reproduction

1. Human Reproductive Organs

- • Male System: Scrotum housing paired Testes (operating at 2-3°C below core body temperature for spermatogenesis), Epididymis, Vas deferens, Seminal vesicles, Prostate gland, Cowper's glands, and Urethra.
- • Female System: Paired Ovaries, Fallopian tubes (Oviducts - site of fertilization), Uterus (muscular womb for blastocyst implantation), Cervix, and Vagina.

2. Hormonal Regulation of the Menstrual Cycle (28-day cycle)

- • Follicular Phase (Days 1–13): Pituitary FSH stimulates primary follicle growth; maturing follicle secretes Estrogen rebuilding endometrium.
- • Ovulation Phase (Day 14): Pituitary LH surge triggers rupture of Graafian follicle, releasing secondary oocyte into oviduct.
- • Luteal Phase (Days 15–28): Ruptured follicle transforms into Corpus Luteum, secreting high levels of Progesterone for uterine lining maintenance.
- • Menstruation (Days 1–5): If unfertilized, Corpus Luteum degenerates into Corpus Albicans; progesterone plummets, shedding endometrium.

3. Assisted Reproductive Technologies (ART)

- • In-Vitro Fertilization (IVF): Oocytes and sperms fertilized in vitro, followed by blastocyst embryo transfer.
- • Surrogacy: Biological embryo implanted into the surrogate mother's uterus for full gestation.
- • Sperm Banks: Cryopreserved donor semen compliant with ethical and infectious disease screening.

UNIT IV: ECOLOGY, BIODIVERSITY & WASTE MANAGEMENT

Ecological Balance, Hotspots, Sacred Groves & 3R Waste Paradigm

1. Biodiversity Levels & Global Hotspots

- • Genetic Diversity: Intra-species variation (e.g. thousands of distinct rice or mango cultivars).
- • Species Diversity: Inter-species richness within a given geographical biome.
- • Ecosystem Diversity: Landscape richness spanning terrestrial, aquatic, estuarine, and marine biomes.
- • Biodiversity Hotspots: Western Ghats, Eastern Himalayas, and Indo-Burma regions housing threatened endemic wildlife.

2. Sacred Groves (Devrai)

- • Ancient forest sanctuaries protected by rural indigenous communities in the name of local deities.
- • Harbor rare medicinal plants, undisturbed seed gene banks, and endemic micro-fauna safe from commercial deforestation.

3. Modern Solid Waste Management Principles

- • Source Segregation: Immediate division into biodegradable wet waste, recyclable dry waste, and domestic hazardous waste.
- • The 3R Philosophy: Reduce consumption, Reuse non-degradable containers, and Recycle plastics, paper, and glass.
- • Vermicomposting: Earthworm cultures (Eisenia fetida) transforming organic vegetable waste into nutrient-rich humus.
- • Sanitary Landfilling: Engineered clay-lined subterranean pits with leachate capture and methane extraction pipes.

UNIT V: TOWARDS GREEN ENERGY SYSTEMS

Thermodynamics of Power Generation: Thermal vs Solar vs Hydroelectric

1. Electromagnetic Induction & Power Generation

- • Discovered by Michael Faraday: When magnetic flux linked with a conductive coil changes, an electric potential difference is induced.
- • All turbine-driven generating stations (thermal, nuclear, hydro, wind) operate on this principle.

2. Comparative Technology Matrix

- • Thermal Power: Coal combustion produces steam driving turbines. Efficiency: ~35%. Issues: Airborne particulate matter (SPM), CO_2 emissions causing global warming, acid rain.
- • Nuclear Power: Fission of Uranium-235 splits atoms releasing massive heat. Zero greenhouse gas emission during operation, but high risk of ionizing radiation leaks and millenia-long toxic nuclear waste disposal.
- • Hydroelectric Power: High-elevation water rotates Kaplan or Pelton turbines. Clean renewable source, but alters aquatic ecosystems and requires massive population resettlement.
- • Solar Photovoltaic (PV): Directly converts photon energy into Direct Current (DC) electricity via semiconductor p-n junctions without mechanical friction or water consumption.

UNIT VI: SOCIAL HEALTH & DIGITAL WELLBEING

Substance Abuse Hazards, Mental Stress Factors & Remedial Coping Strategies

1. Dimensions of Social Health

- • Social health is the capacity of an individual to establish harmonious, empathetic, and constructive interpersonal relationships within community and family structures.
- • Threats to social health: Chronic diseases, poverty, loneliness, communal prejudices, addiction, and uncontrolled cyber obsession.

2. Addiction Hazards (Tobacco & Alcohol)

- • Tobacco & Gutkha: Contain highly toxic carcinogens (nicotine, tar, nitrosamines) inducing oral cancer, emphysema, and cardiovascular ischemia.
- • Alcohol: Causes hepatic cirrhosis, neurodegenerative cognitive decline, delayed reflexes, and economic devastation of households.

3. Cyber Threats & Stress Management Protocols

- • Excessive Smartphone & Gaming Use: Impairs sleep melatonin cycles, fosters social withdrawal, and risks cyber exploitation.
- • Stress Alleviation Measures: Physical sports, musical arts, laughter therapy clubs, nature interaction, family communication, and statutory helpline support (Childline 1098 / Tele-MANAS).

UNIT VII: DISASTER MANAGEMENT PROTOCOLS

Disaster Classification, Mitigation Lifecycle & First Aid Action Principles

1. Disaster Classification

- • Natural Disasters: Earthquakes, tsunamis, volcanic eruptions, tropical cyclones, cloudburst flash floods, droughts.
- • Man-made Disasters: Industrial chemical gas leaks (Bhopal Disaster 1984), nuclear radiation accidents (Chernobyl 1986, Fukushima 2011), war, and terrorism.

2. Disaster Mitigation Lifecycle

- • 1. Pre-disaster Preparation: Risk hazard mapping, civil defense training, and disaster mock drills in educational centers.
- • 2. Warning Phase: Sirens, radio broadcasts, and mandatory coastal evacuations.
- • 3. Emergency Rescue Phase: Search, extraction, medical triage by NDRF (National Disaster Response Force).
- • 4. Rehabilitation Phase: Debris clearance, safe potable water, epidemic vaccination, transient shelters.
- • 5. Reconstruction Phase: Seismic-resistant architectural construction and economic livelihood revival.

3. Vital First Aid Action Protocols

- • ABC Protocol: Airway clearance -> Breathing verification -> Circulation check and chest compressions (CPR).
- • Severe Hemorrhage: Apply direct pressure with sterile pad; apply tourniquet proximal to wound if arterial bleeding persists.
- • Burn Injuries: Irrigate with clean cool flowing water for at least 15 minutes; do not apply unverified oily ointments.

SOLVED EXEMPLARS 1: OBJECTIVES & FACTUAL RECALL

Model 1-Mark Questions Solved with Board Marking Rubrics | Bloom: Remember

Factual & Objective Model Questions

- Q1. Complete the statement: The enzyme responsible for synthesizing messenger RNA from DNA template is _____.
- *Model Answer: RNA polymerase. (Mark: 1)*
- Q2. Identify correlation: Connecting link between Reptiles and Mammals : Duck-billed platypus :: Connecting link between Annelida and Arthropoda : _____.
- *Model Answer: Peripatus. (Mark: 1)*
- Q3. Which scientist first proposed the concept of "Survival of the Fittest"?
- *Model Answer: Herbert Spencer coined the phrase, and Charles Darwin adopted it in "Origin of Species" to describe natural selection. (Mark: 1)*
- Q4. Find the odd one out: Appendix, Wisdom teeth, Eye lens, Coccyx.
- *Model Answer: Eye lens. Reason: Eye lens is an active, essential functional optical organ, while appendix, wisdom teeth, and coccyx are vestigial organs in humans. (Mark: 1)*
- Q5. Name the bacteria engineered to biologically decompose maritime petroleum oil spills.
- *Model Answer: Alcanivorax borkumensis (Hydrocarbonoclastic bacteria). (Mark: 1)*
- Q6. True or False: "Solar thermal electricity generation directly converts sunlight to DC using semiconductor silicon wafers."
- *Model Answer: False. Solar photovoltaic (PV) cells directly convert sunlight to DC; solar thermal plants use parabolic reflectors to boil water and drive steam turbines. (Mark: 1)*

SOLVED EXEMPLARS 2: SCIENTIFIC REASONS & 2-MARK ANSWERS

Conceptual Justifications with Logical Scientific Cause-and-Effect Sequences

Scientific Reasoning Model Answers

- Q1. Give scientific reason: The study of developing embryos provides compelling evidence for organic evolution. (2 Marks)
- *Model Answer: (1) Comparative embryological examinations reveal that very young embryos of diverse vertebrates (fish, salamanders, tortoises, birds, humans) exhibit strikingly identical morphological characteristics, including pharyngeal gill clefts and rudimentary tail structures.*
- (2) As developmental gestation proceeds, specialized class distinctions emerge. This initial embryonic homology proves that all vertebrate classes evolved from a common evolutionary ancestor. (Mark: 2)
- Q2. Give scientific reason: Microbial enzymes are replacing traditional chemical catalysts in modern industrial chemistry. (2 Marks)
- *Model Answer: (1) Microbial enzymes operate efficiently at ambient temperature, normal atmospheric pressure, and neutral pH, resulting in immense thermal energy savings compared to chemical processes.*
- (2) Furthermore, enzymes exhibit absolute substrate specificity, eliminate hazardous heavy-metal pollutants, and produce minimal toxic chemical effluents. (Mark: 2)
- Q3. Give scientific reason: The sex of an unborn human infant is determined strictly by the paternal contribution. (2 Marks)
- *Model Answer: (1) Human females possess homogeneous sex chromosomes (44 + XX) and produce only X-bearing ova.*
- (2) Human males are heterogametic (44 + XY) producing 50% X-bearing sperms and 50% Y-bearing sperms. Fertilization of an ovum with a Y-sperm yields a male (XY), while fertilization with an X-sperm yields a female (XX). Hence, father’s sperm alone determines gender. (Mark: 2)

SOLVED EXEMPLARS 3: COMPARATIVE DISTINCTIONS & TABLES

Structured Tabular Comparisons for High-Yield Board Marks (2 & 3 Marks)

Comparison 1: Darwinism vs Lamarckism

- • Point of Difference: Basis of Evolutionary Change
- Darwinism: Natural selection acts upon existing inherent genetic variations within a population.
- Lamarckism: Morphological changes occur due to deliberate use or disuse of organs during an individual’s lifetime.
- • Point of Difference: Heritability of Modifications
- Darwinism: Only germline genetic mutations are transmitted to successive offspring.
- Lamarckism: Somatic acquired characters are claimed to be inherited (a claim scientifically disproven).
- • Point of Difference: Published Milestone
- Darwinism: "Origin of Species" (1859). | Lamarckism: "Philosophie Zoologique" (1809).

Comparison 2: Solar Photovoltaic vs Conventional Thermal Electricity

- • Fuel & Energy Source: Solar PV uses boundless solar photon radiation; Thermal uses finite fossil coal.
- • Environmental Byproducts: Solar PV has zero greenhouse emissions during operation; Thermal releases CO2, SO2, and fly ash.
- • Mechanical Operation: Solar PV has no moving machinery (solid-state semiconductor); Thermal uses high-pressure steam boilers and rotating generators.

SOLVED EXEMPLARS 4: 3-MARK ANALYTICAL QUESTIONS

Bloom: Evaluate & Analyze | Methodical Multi-Point Explanations

Analytical Questions & Justifications

- Q1. Read the statement and justify in your own words: "Geographical and reproductive isolation of organisms gradually leads to speciation." (3 Marks)
- *Model Answer: (1) Speciation is the evolutionary process of new species formation. Each natural species is uniquely adapted to specific geographical habitats, nutritional niches, and reproductive seasons.*
- (2) When physical geological barriers (e.g., canyon formation, rising sea levels, mountain building) divide a parent population, interbreeding between sub-populations halts entirely (Geographical Isolation).
- (3) Over prolonged generations, each isolated sub-population undergoes independent spontaneous mutations and experiences distinct environmental selection forces.
- (4) Eventually, anatomical or chromosomal divergences prevent them from successfully interbreeding even if they reunite (Reproductive Isolation). This culmination establishes separate, permanent species. (Mark: 3)
- Q2. Explain why conducting disaster management mock drills in educational institutions is of vital importance. (3 Marks)
- *Model Answer: (1) Mock drills evaluate actual institutional response speed, evacuation efficiency, and operational coordination during simulated fires or earthquakes.*
- (2) They verify whether emergency fire extinguishers, alarm systems, and exit corridors are clear and fully functional.
- (3) They condition pupils and faculty to eliminate panic, adopt the Drop-Cover-Hold safety posture, and practice essential rescue and bandage skills. (Mark: 3)

SOLVED EXEMPLARS 5: 5-MARK LONG ANSWER MASTERY

Comprehensive Essays, Multi-Step Biological Mechanisms & Flowcharts

Q. Detail the complete molecular pathway of Protein Synthesis (Central Dogma) including Transcription, T

- Model Answer Architecture:
- 1. Conceptual Definition: Proteins dictate cellular biochemical processes. The synthesis of functional proteins directed by DNA through intermediary RNA molecules is termed Central Dogma.
- 2. Transcription: Inside the nucleus, RNA polymerase attaches to the DNA promoter. The complementary single strand of mRNA is formed against the template strand. Adenine pairs with Uracil (instead of Thymine), and Cytosine pairs with Guanine. After synthesis, mRNA translocates through nuclear pores into the cytoplasm. (Mark: 1.5)
- 3. Translation: The mRNA binds to a ribosome. Each three successive nitrogenous bases represent a Triplet Codon specifying an amino acid. Specific transfer RNA (tRNA) possessing an Anticodon complementary to the mRNA codon delivers the corresponding amino acid. (Mark: 1.5)
- 4. Translocation & Peptide Bonding: The ribosome moves progressively along the mRNA strand by the distance of exactly one codon. Ribosomal RNA (rRNA) catalytically forms peptide bonds between consecutive amino acids, elongating the polypeptide chain until a stop codon (UAA, UAG, UGA) is reached. The protein then folds into its active tertiary conformation. (Mark: 2)

Q. Describe Flood Disaster Management across Pre-Disaster, Emergency, and Rehabilitation Phases. (5 M

- Model Answer Highlights: Pre-disaster involves flood-plain zoning, river dredging, and meteorological radar alerts. Emergency involves NDRF motorized boat rescue, medical triage, and food drop operations. Rehabilitation involves water chlorination against cholera epidemics, structural repairs of bridges, and economic compensation for farmers.

HIGH-YIELD REVISION MATRIX & EXAM STRATEGY

Final Board Examination Readiness Checklist & Golden Rules for 40/40

Science 2 Golden Examination Rules

- 1. Diagram Rule: In Biology (Human Reproductive System, Central Dogma, Mitosis), draw neat pencil diagrams with all labels on the right-hand side using a ruler.
- 2. Keywords Rule: Bold or underline biological terms: Transcription, Uracil, Anticodon, Homologous, Vestigial, NDRF, Photovoltaic.
- 3. Time Allocation Strategy (120 Minutes Total):
  - Q.1 (A & B) [10 Marks]: 20 Minutes (Read carefully, avoid guessing without elimination).
  - Q.2 (A & B) [10 Marks]: 25 Minutes (Bullet points with scientific cause and effect).
  - Q.3 [15 Marks]: 45 Minutes (Provide 3 distinct substantial analytical points for every 3-mark question).
  - Q.4 [5 Marks]: 20 Minutes (Structured long answer with headings and diagrams).
  - Final Review: 10 Minutes (Check question numbers, diagrams, and units).

Mastery Checklist across 22 Science 2 Dataset Topics

Ensure complete mastery of: heredity, human evolution, theories of evolution, evidences of evolution, protein synthesis, dna genes and chromosomes, mutation, types of disasters, disaster management cycle, first aid, disaster signs, human reproductive system, reproductive health, biodiversity, solar energy, addiction, interpersonal relationships, social media, stress management, and superstitions.